

Reparametrizations

As discussed the last time, we must keep in mind that parametrizations are means to an end

Reparametrizations

There are many parametrizations of the same curve

Reparametrizations

and we have to ensure that changing the parametrization does not change the concept

Reparametrizations

$$\theta : (a, b) \rightarrow \mathbb{R}^2$$

So let us consider some parametrization of a curve

Reparametrizations

$$\theta : (a, b) \rightarrow \mathbb{R}^2$$

$$\theta_1(t) = (\cos(t), \sin(t))$$

For example, let us consider the circle

Reparametrizations

$$\theta : (a, b) \rightarrow \mathbb{R}^2$$

$$\theta_1(t) = (\cos(t), \sin(t))$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

As we saw the last time, here is another among many ways to parametrize it

Reparametrizations

$$\theta : (a, b) \rightarrow \mathbb{R}^2$$

$$\theta_1(t) = (\cos(t), \sin(t)) \text{ where } t \in (0, 2\pi)$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

So in this case, if we needed the interval $(0, 2\pi)$

Reparametrizations

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Since the second one is at twice the speed, you need half the interval

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$$(0, 2\pi) \xrightarrow{\theta_1} \mathbb{R}^2$$

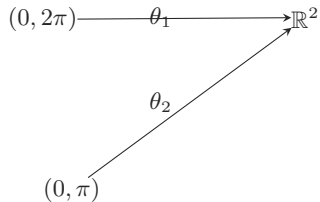
θ_1 is “drawing” one way to draw the circle

Reparametrizations

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and θ_2 is another way. But how are they related?

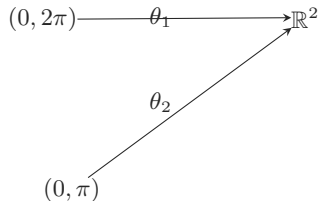
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Notice that they are related by the highlighted expression

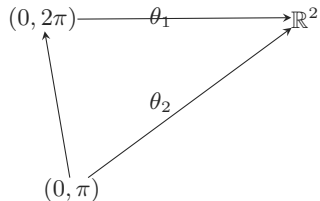
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In other words, we have a map between the intervals

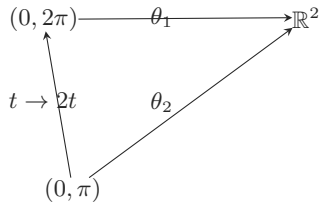
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Which is t going to $2t$

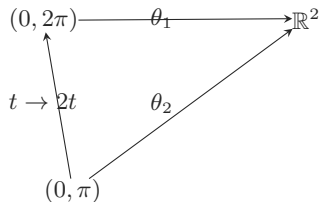
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Consider two parametrizations

We can generalize this to any parametrizations

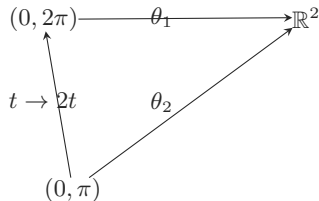
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Consider two parametrizations

$$\theta_1 : (a, b) \rightarrow \mathbb{R}^2$$

If we represent the curve by some parametrization

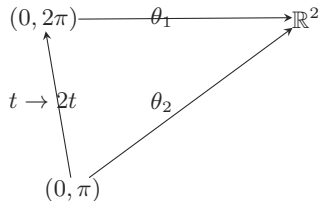
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Consider two parametrizations

$$\theta_1 : (a, b) \rightarrow \mathbb{R}^2$$

$$\theta_2 : (c, d) \rightarrow \mathbb{R}^2$$

and then choose an alternative parametrization

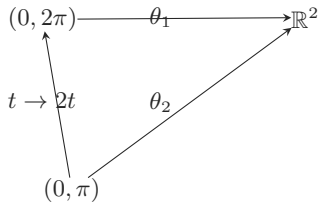
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Consider two parametrizations

$$\theta_1 : (a, b) \rightarrow \mathbb{R}^2$$

$$\theta_2 : (c, d) \rightarrow \mathbb{R}^2$$

, then a reparametrization $\tau : (c, d) \rightarrow (a, b)$

That function $2t$ can be generalized to τ

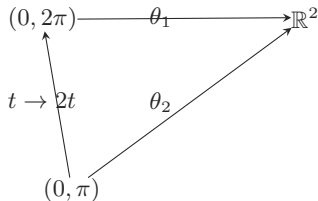
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$$\theta_1(t) = \theta_2(\tau(t))$$

Let us recall the definition of curvature

Reparametrizations

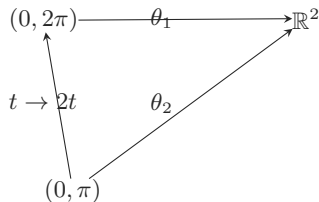
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Given a curve,



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of a given curve...

Reparametrizations

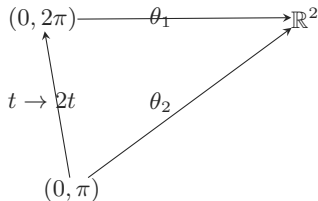
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Given a curve, find a unit speed parametrization, θ



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which we represent by some parametrization, but this time we have to ensure it is unit speed

Reparametrizations

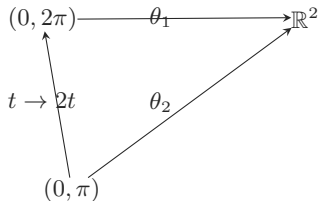
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Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$)



Consider two parametrizations

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, then a reparametrization $\tau : (c, d) \rightarrow (a, b)$

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which means that the magnitude of the velocity should be 1 for any t

Reparametrizations

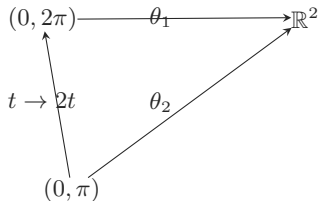
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Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$



Consider two parametrizations

$$\theta_1 : (a, b) \rightarrow \mathbb{R}^2$$

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, then a reparametrization $\tau : (c, d) \rightarrow (a, b)$

$$\theta_1(t) = \theta_2(\tau(t))$$

Now that it is unit speed, any acceleration is owing to a change in direction

Reparametrizations

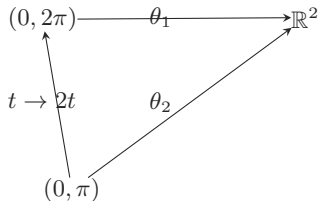
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so it is a good measure of the curvature

Reparametrizations

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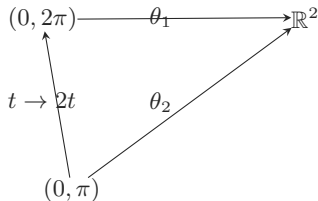
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Example



Consider two parametrizations

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To get used to this definition let us try it with an example

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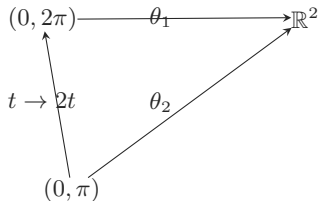
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Example

$$\theta(t) = (\cos(t), \sin(t))$$



Consider two parametrizations

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Let us take the circle, which we are parametrizing by this

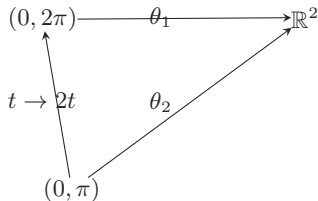
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Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$

Let us consider some point

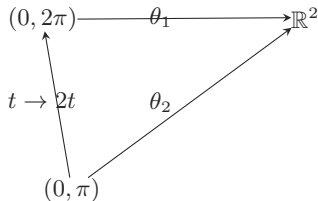
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Example

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Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle

which we know lies on the circle

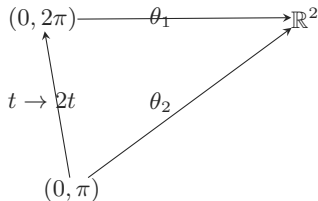
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Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

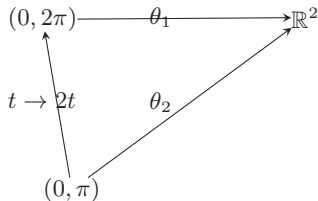
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Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

We compute its velocity by taking the derivative

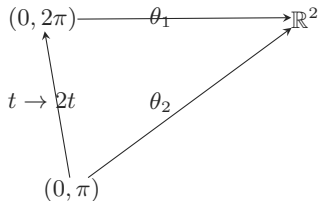
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Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2}$$

To calculate its magnitude, we take its "norm"

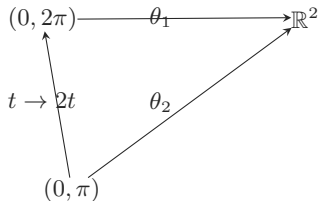
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Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

and it is equal to 1 in this case so it is unit speed

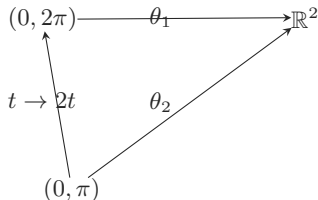
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$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| =$$

We are, therefore, ready to compute the magnitude of the acceleration

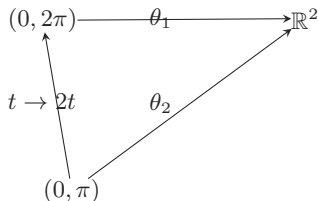
Reparametrizations

$$\theta : (a, b) \rightarrow \mathbb{R}^2$$

$$\theta_1(t) = (\cos(t), \sin(t)) \text{ where } t \in (0, 2\pi)$$

$$\theta_2(t) = (\cos(2t), \sin(2t)) \text{ where } t \in (0, \pi)$$

$$\theta_2(t) = \theta_1(2t)$$



Consider two parametrizations

$$\theta_1 : (a, b) \rightarrow \mathbb{R}^2$$

$$\theta_2 : (c, d) \rightarrow \mathbb{R}^2$$

, then a reparametrization $\tau : (c, d) \rightarrow (a, b)$

$$\theta_1(t) = \theta_2(\tau(t))$$

Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

which you can check is also 1

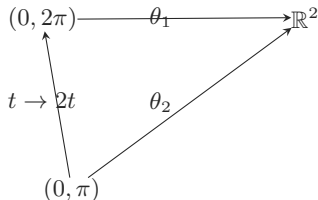
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Consider two parametrizations

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Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

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Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

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$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

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$$\theta_2(t) = (\cos(2t), \sin(2t))$$

But what if we used this different parametrization instead?

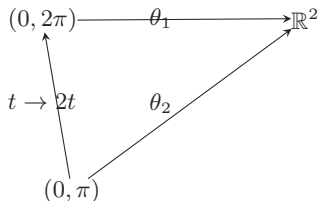
Reparametrizations

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$$\theta_2(t) = (\cos(2t), \sin(2t)) \text{ where } t \in (0, \pi)$$

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Consider two parametrizations

$$\theta_1 : (a, b) \rightarrow \mathbb{R}^2$$

$$\theta_2 : (c, d) \rightarrow \mathbb{R}^2$$

, then a reparametrization $\tau : (c, d) \rightarrow (a, b)$

$$\theta_1(t) = \theta_2(\tau(t))$$

Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$

Once again, we consider the same point

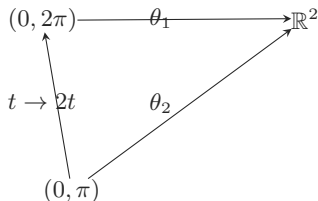
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Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

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$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle

which, remember, lies on the circle of radius 1, but its parameter will be different

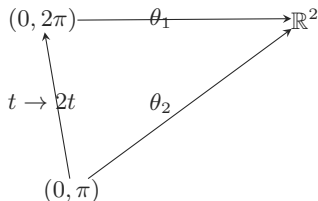
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Consider two parametrizations

$$\theta_1 : (a, b) \rightarrow \mathbb{R}^2$$

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, then a reparametrization $\tau : (c, d) \rightarrow (a, b)$

$$\theta_1(t) = \theta_2(\tau(t))$$

Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

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$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

This time it is an image of $\pi/8$ since we are using θ_2 now

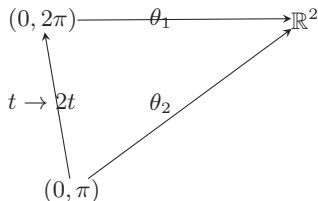
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Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

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$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

By simple chain rule, its velocity will be as highlighted

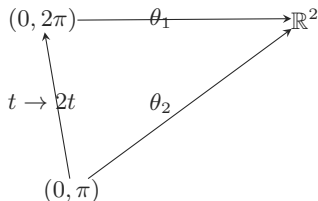
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Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

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Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

This time, if you compute the magnitude it will be 2, so it is constant, but not unit speed

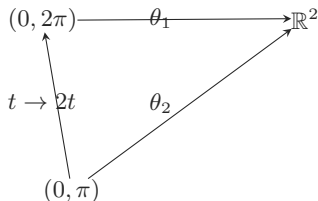
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$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

$$\theta(t) = \theta_2(t/2)$$

But we can change it into a unit speed one by first halving the t

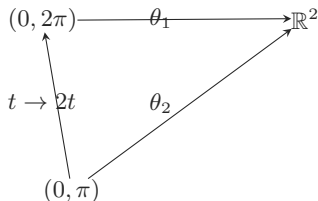
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$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

$$\theta(t) = \theta_2(t/2)$$

in which case it will be the previous parametrization and we can compute the curvature as before

Can every parametrization be changed to a unit speed one?

Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

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$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

$$\theta(t) = \theta_2(t/2)$$

Can we do this for any curve?

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

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Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

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$$\|\theta'(t)\| = 2$$

$$\theta(t) = \theta_2(t/2)$$

Supposing we take such a complicated parametrization as shown

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

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$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

$$\theta(t) = \theta_2(t/2)$$

it is not hard to guess the reparametrization necessary to make it unit speed

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization

Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

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Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

$$\theta(t) = \theta_2(t/2)$$

But unless we do so, we will find it very difficult to find a formula for the curvature

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization

Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

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Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

$$\theta(t) = \theta_2(t/2)$$

that will work with any parametrization

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization

Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

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Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

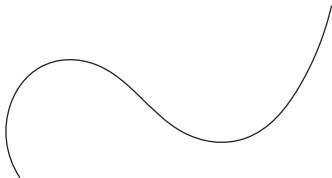
$$\theta(t) = \theta_2(t/2)$$

So we need to answer: can we always find a unit speed reparametrization of any given parametrization?

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

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Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

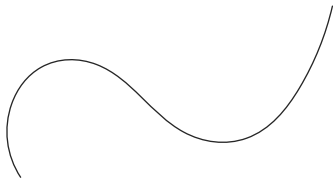
$$\theta(t) = \theta_2(t/2)$$

Consider a general curve as shown

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

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$$\theta(t) = (\cos(t), \sin(t))$$

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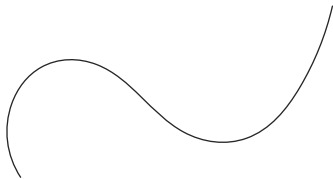
$$\theta(t) = \theta_2(t/2)$$

And let us consider a parametrization of it from some interval (which is shown by the line)

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

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$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

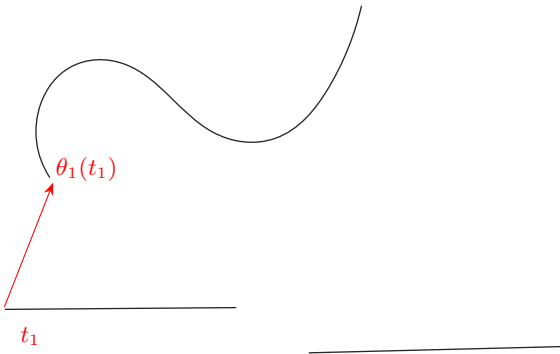
$$\theta(t) = \theta_2(t/2)$$

I could also parametrize it using a completely different parametrization with its own interval

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

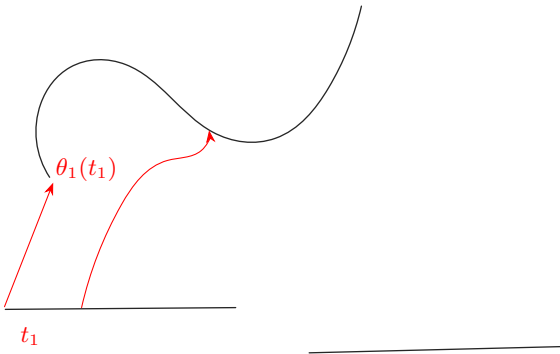
$$\theta(t) = \theta_2(t/2)$$

The beginning of the first interval is sent to the beginning of the curve

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

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$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

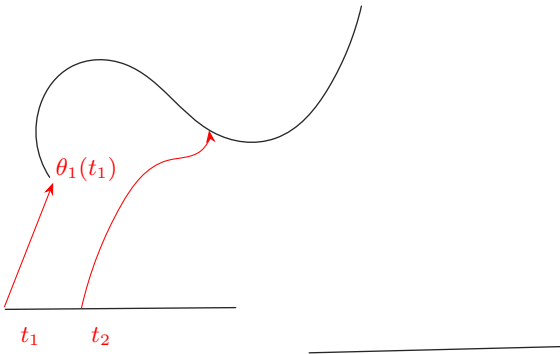
$$\|\theta'(t)\| = 2$$

$$\theta(t) = \theta_2(t/2)$$

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

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$$\theta(t) = (\cos(t), \sin(t))$$

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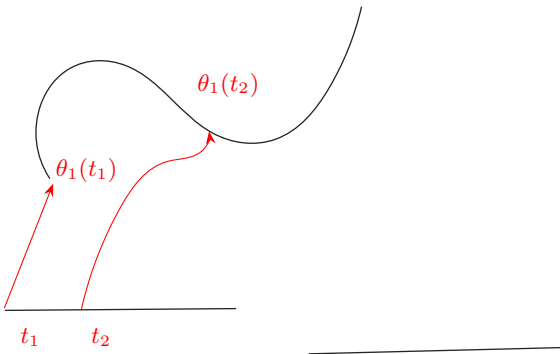
Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

$$\theta(t) = \theta_2(t/2)$$

Goal: Find a definition of curvature that can use any parametrization



Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2 \sin(2t), 2 \cos(2t))$$

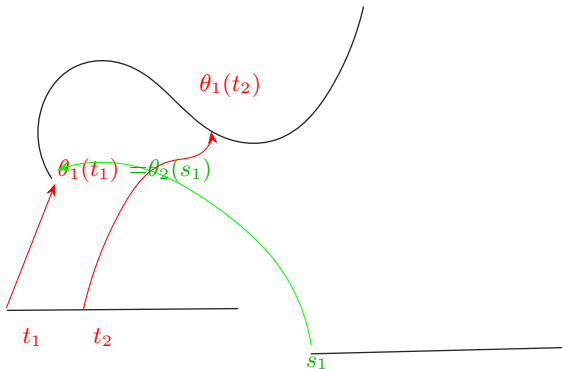
$$\|\theta'(t)\| = 2$$

$$\theta(t) = \theta_2(t/2)$$

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

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$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

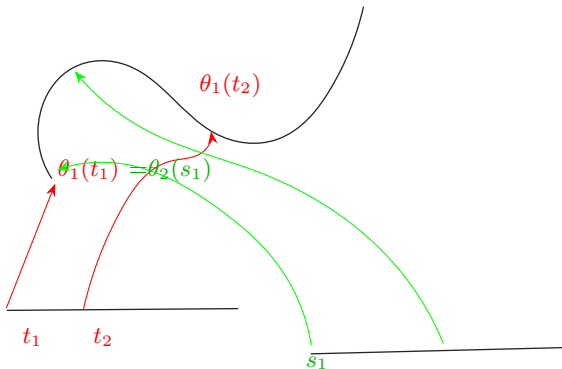
$$\theta(t) = \theta_2(t/2)$$

The second parametrization too sends the beginning of the interval to the beginning of the curve

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

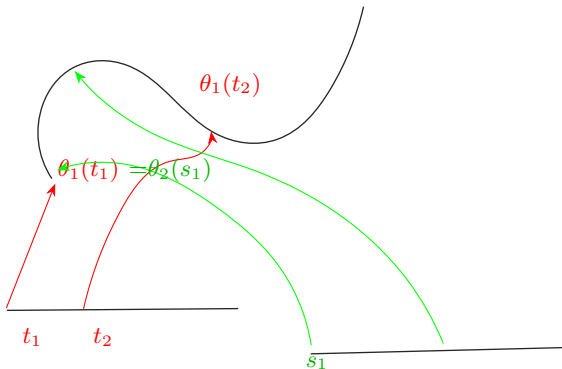
$$\theta(t) = \theta_2(t/2)$$

But this time we send a parameter s_2 to a point

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

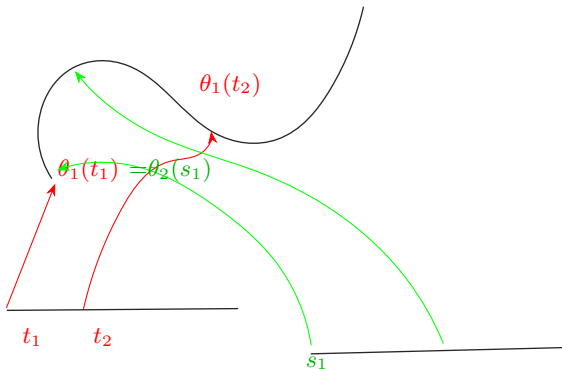
$$\theta(t) = \theta_2(t/2)$$

that is exactly the same distance away from the beginning of the curve

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

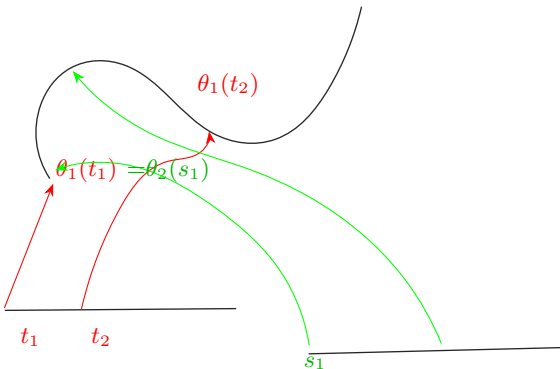
$$\theta(t) = \theta_2(t/2)$$

as s_2 is from the interval to which it belongs

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

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$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

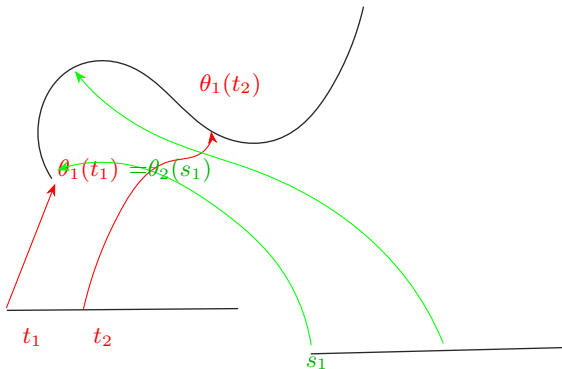
$$\theta(t) = \theta_2(t/2)$$

This is a parametrization for which the “time” parameter coincides with the distance traversed!

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

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$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

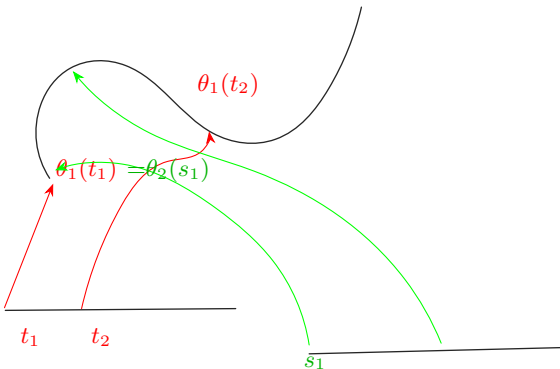
$$\theta(t) = \theta_2(t/2)$$

In other words, it is unit speed!

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

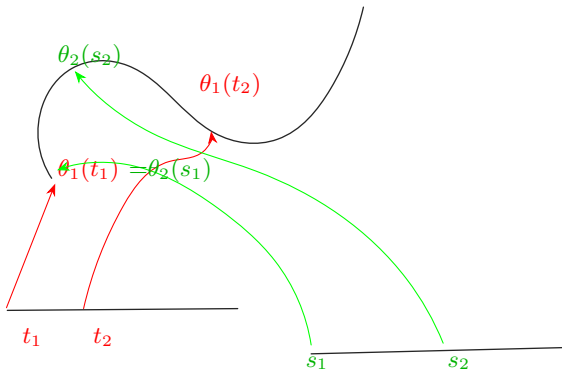
$$\theta(t) = \theta_2(t/2)$$

Unit speed parametrizations are precisely those parametrizations whose parameter corresponding to a point

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

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$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

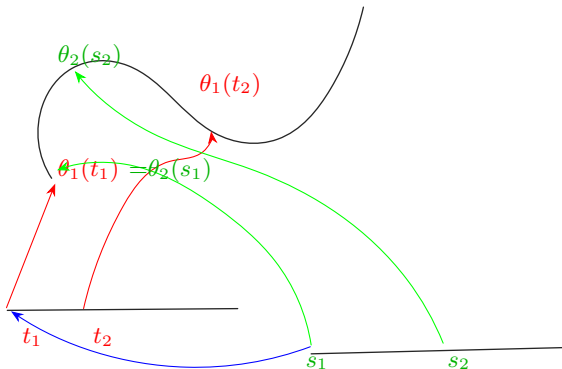
$$\theta(t) = \theta_2(t/2)$$

is exactly the "arc length" of the part of the curve already traversed!

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

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$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

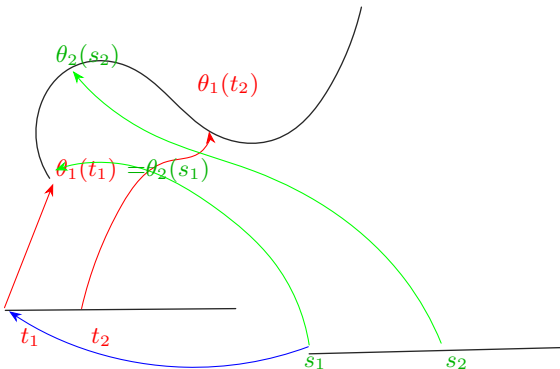
$$\theta(t) = \theta_2(t/2)$$

Now we can see that when switching from parametrization θ_2 to θ_1 we must change s_1 to t_1

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

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$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

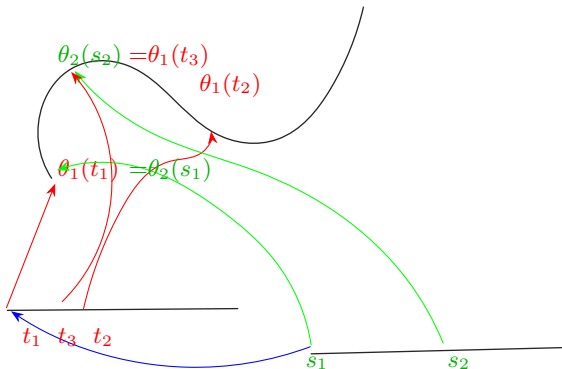
$$\theta(t) = \theta_2(t/2)$$

because s_1 gets taken by θ_2 to exactly the same point as t_1 is taken by θ_1

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

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$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

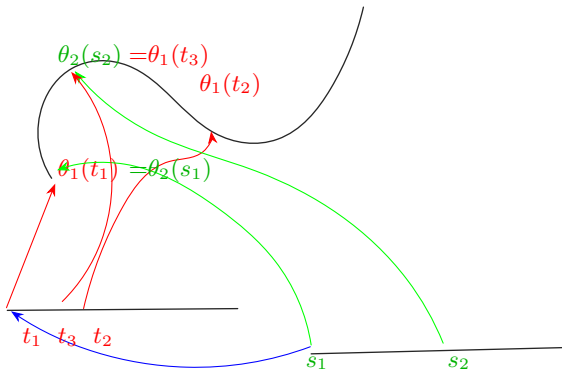
$$\theta(t) = \theta_2(t/2)$$

To figure out what s_2 corresponds to we need to find out what θ_1 sends to $\theta_2(s_2)$

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

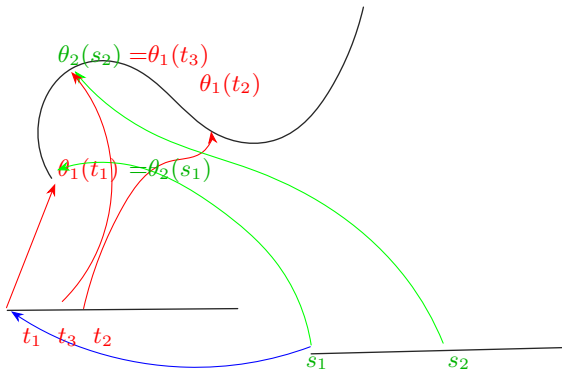
$$\theta(t) = \theta_2(t/2)$$

To find out what s_2 corresponds to, we have to find out what θ_1 sends to the same point

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



We call that parameter t_3

Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

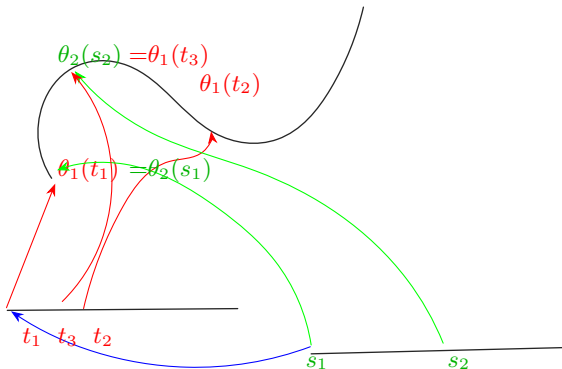
$$\|\theta'(t)\| = 2$$

$$\theta(t) = \theta_2(t/2)$$

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

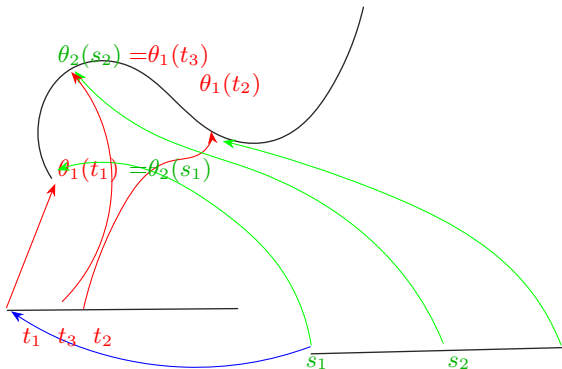
$$\|\theta'(t)\| = 2$$

$$\theta(t) = \theta_2(t/2)$$

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

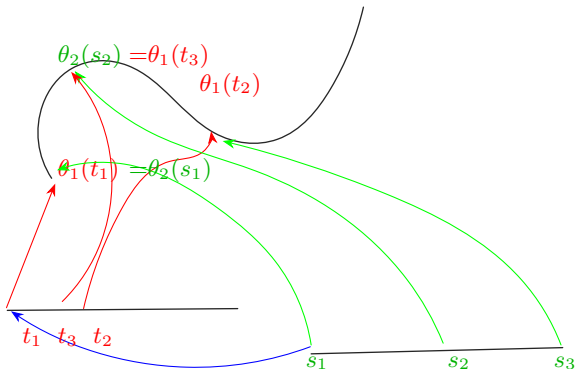
$$\|\theta'(t)\| = 2$$

$$\theta(t) = \theta_2(t/2)$$

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta(\pi/4)$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\|\theta'(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1$$

$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

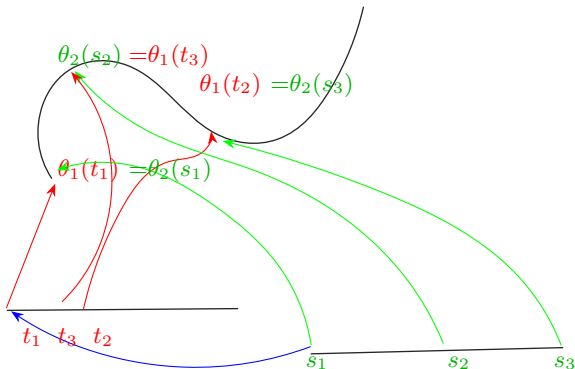
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Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

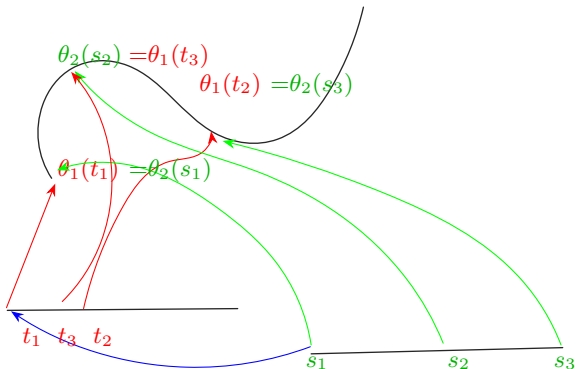
$$\theta(t) = \theta_2(t/2)$$

As before, for any other parameter s_3 for θ_2

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

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$$\|\theta''(\pi/4)\| = 1$$

$$\theta_2(t) = (\cos(2t), \sin(2t))$$

Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

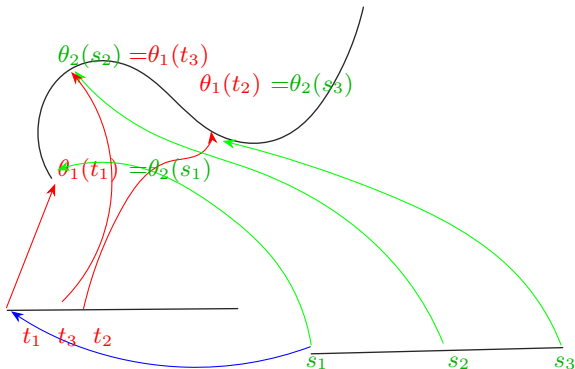
$$\theta(t) = \theta_2(t/2)$$

its such that its distance from the beginning of the interval

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

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Consider the point $(1/\sqrt{2}, 1/\sqrt{2})$ which lies on the circle and is $\theta_2(\pi/8)$

$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

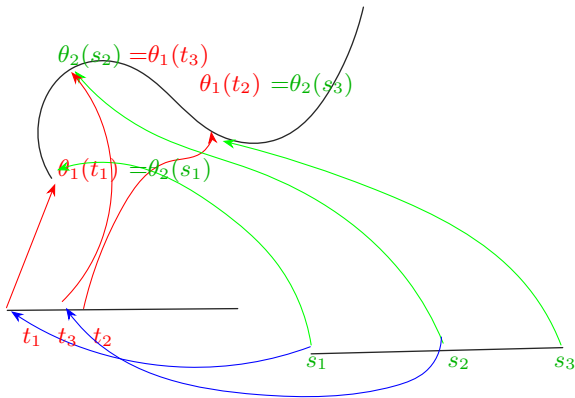
$$\theta(t) = \theta_2(t/2)$$

must be the distance of $\theta_1(t_2)$ from the beginning of the arc

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

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$$\theta'(t) = (-2\sin(2t), 2\cos(2t))$$

$$\|\theta'(t)\| = 2$$

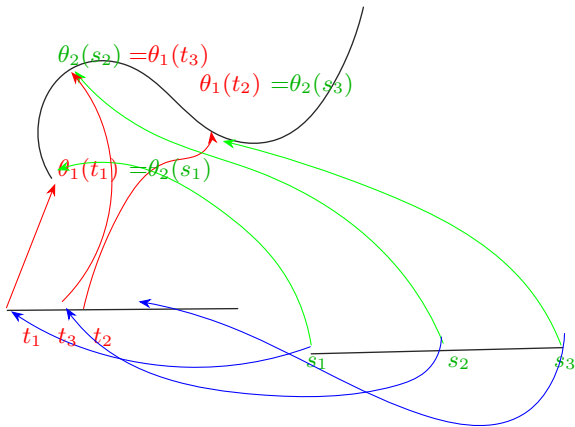
$$\theta(t) = \theta_2(t/2)$$

So now we know what s_2 and s_3 must correspond to

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Given a curve, find a unit speed parametrization, θ (i.e. $\|\theta'(t)\| = 1$), then the curvature at the point $\theta(t)$ is $\|\theta''(t)\|$

Example

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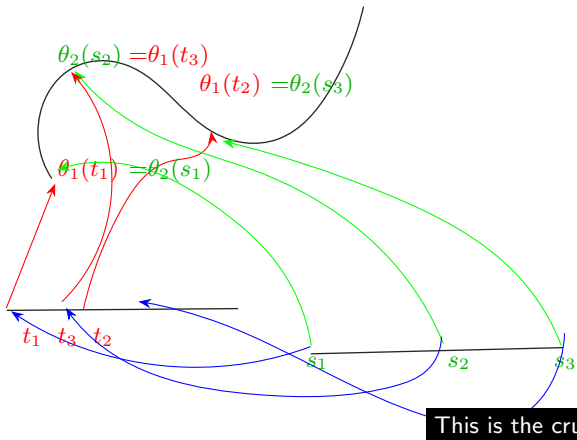
$$\theta(t) = \theta_2(t/2)$$

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization

The parameter in a unit speed parametrization coincides with the arc length

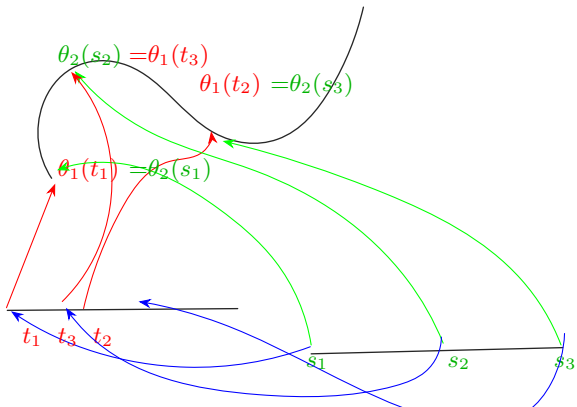


Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization

The parameter in a unit speed parametrization coincides with the arc length



In a unit speed parametrization, the parameter corresponds to the arc length traversed until then

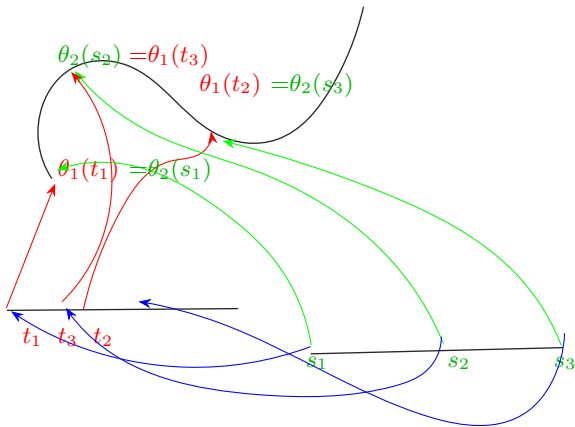
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Goal: Find a definition of curvature that can use any parametrization

The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)



Can every parametrization be changed to a unit speed one?

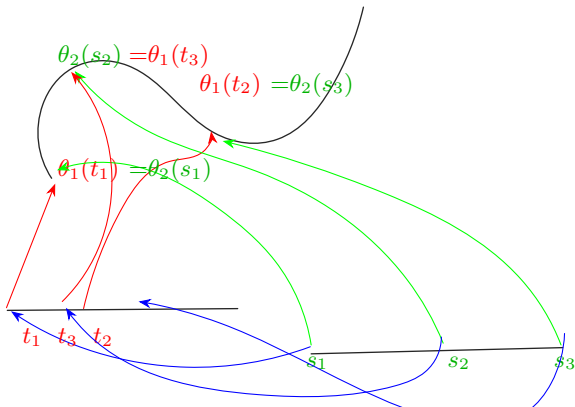
$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization

The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

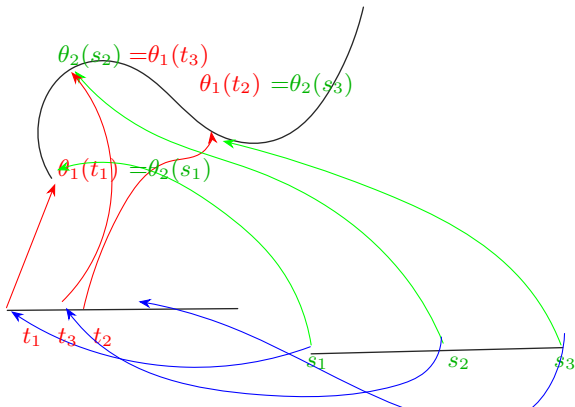


This makes it easy to convert (theoretically) to a unit speed parametrization

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

If we are given a non-unit speed parametrization θ

because for any non unit speed parametrization

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization

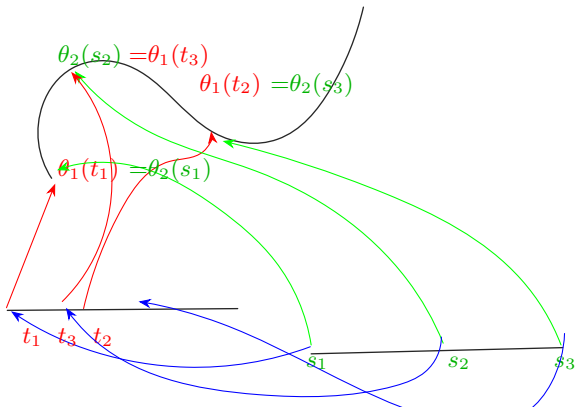
The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

If we are given a non-unit speed parametrization θ

Consider a point p and $p = \theta(t_1) = \theta_2(t_2)$



Any point on it, is the image of of some t_1 under the non-unit speed parametrization θ

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization

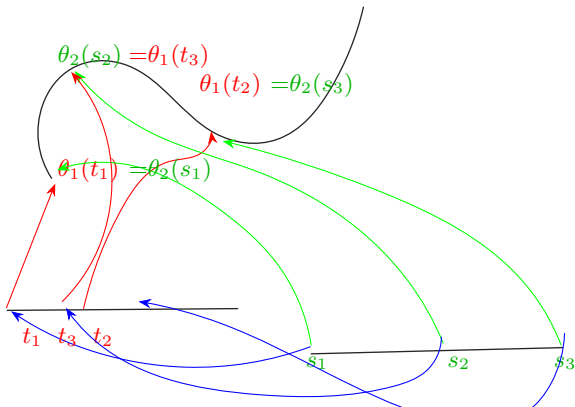
The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

If we are given a non-unit speed parametrization θ

Consider a point p and $p = \theta(t_1) = \theta_2(t_2)$

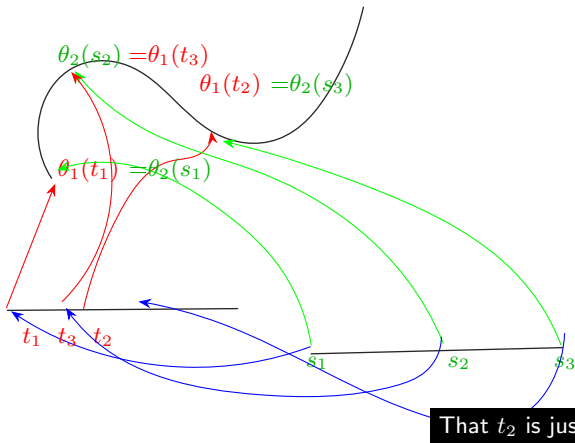


But if it is the image under the unit speed parametrization of, say, t_2

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

If we are given a non-unit speed parametrization θ

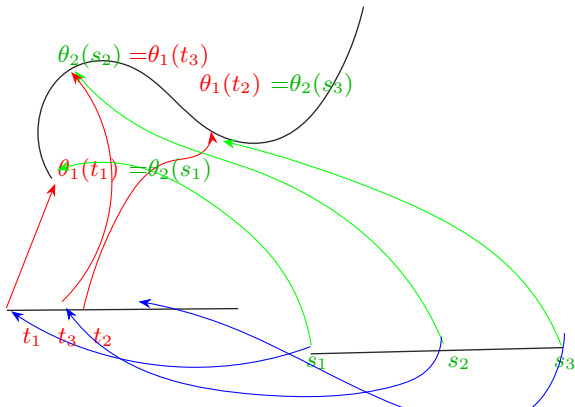
Consider a point p and $p = \theta(t_1) = \theta_2(t_2)$

t_2 is the arc length of the curve until $p = \theta(t_1)$

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

If we are given a non-unit speed parametrization θ

Consider a point p and $p = \theta(t_1) = \theta_2(t_2)$

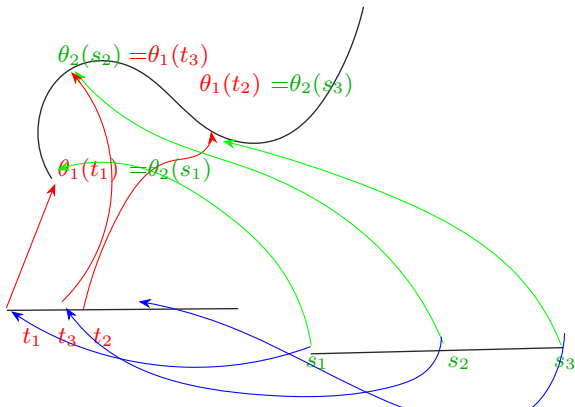
t_2 is the arc length of the curve until $p = \theta(t_1)$

can we figure out the arc length using only θ ?

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



That raises this important question

The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

If we are given a non-unit speed parametrization θ

Consider a point p and $p = \theta(t_1) = \theta_2(t_2)$

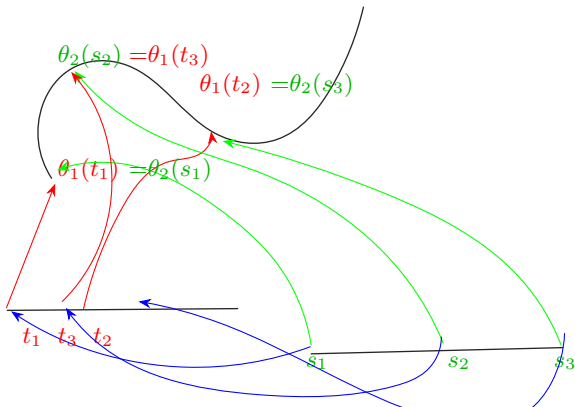
t_2 is the arc length of the curve until $p = \theta(t_1)$

How would you define the arc length of the curve between the points $\theta(t_1)$ and $\theta(t_2)$

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



The trouble is that a curve is curved

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If we are given a non-unit speed parametrization θ

Consider a point p and $p = \theta(t_1) = \theta_2(t_2)$

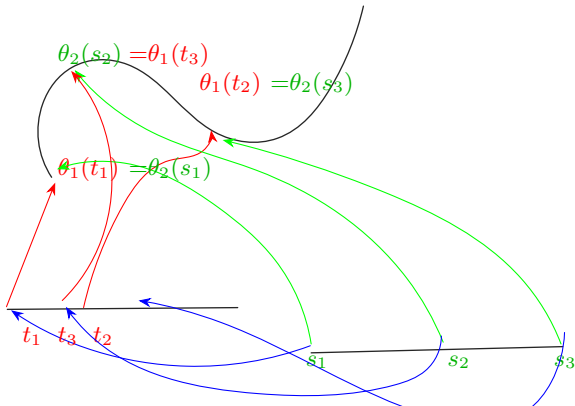
t_2 is the arc length of the curve until $p = \theta(t_1)$

How would you define the arc length of the curve between the points $\theta(t_1)$ and $\theta(t_2)$

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

If we are given a non-unit speed parametrization θ

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t_2 is the arc length of the curve until $p = \theta(t_1)$

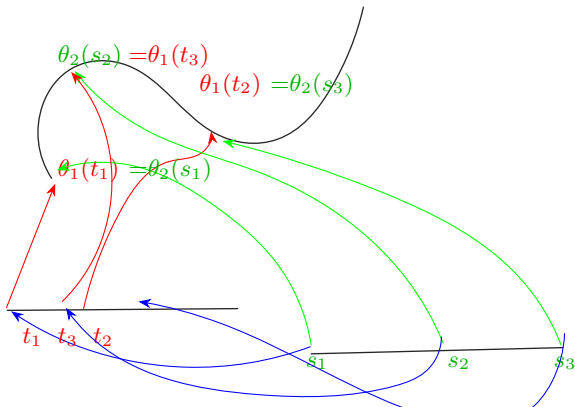
How would you define the arc length of the curve between the points $\theta(t_1)$ and $\theta(t_2)$

so computing distances is more difficult. Remember that we are not talking about displacement.

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

If we are given a non-unit speed parametrization θ

Consider a point p and $p = \theta(t_1) = \theta_2(t_2)$

t_2 is the arc length of the curve until $p = \theta(t_1)$

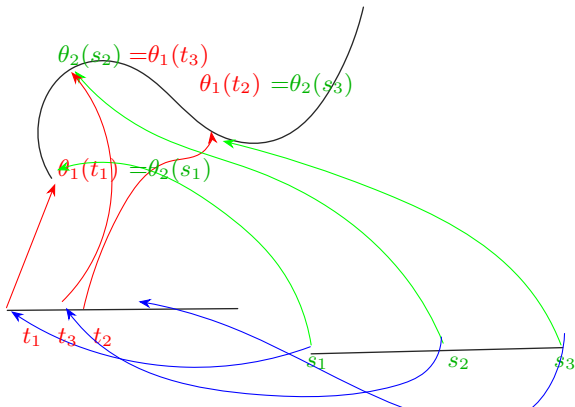
How would you define the arc length of the curve between the points $\theta(t_1)$ and $\theta(t_2)$

However, from the parametrization we can figure out the speed.

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

If we are given a non-unit speed parametrization θ

Consider a point p and $p = \theta(t_1) = \theta_2(t_2)$

t_2 is the arc length of the curve until $p = \theta(t_1)$

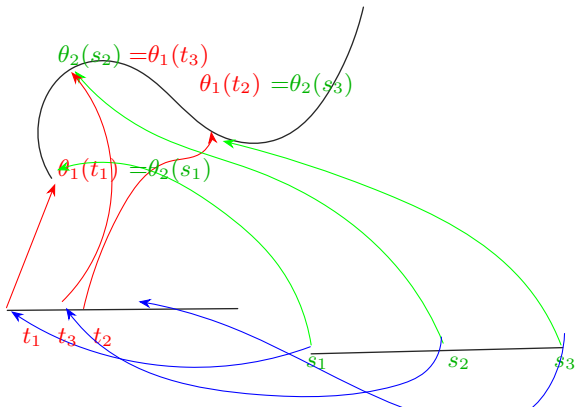
How would you define the arc length of the curve between the points $\theta(t_1)$ and $\theta(t_2)$

Once we know the speed, we can simply integrate to get the distance!!

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

If we are given a non-unit speed parametrization θ

Consider a point p and $p = \theta(t_1) = \theta_2(t_2)$

t_2 is the arc length of the curve until $p = \theta(t_1)$

How would you define the arc length of the curve between the points $\theta(t_1)$ and $\theta(t_2)$

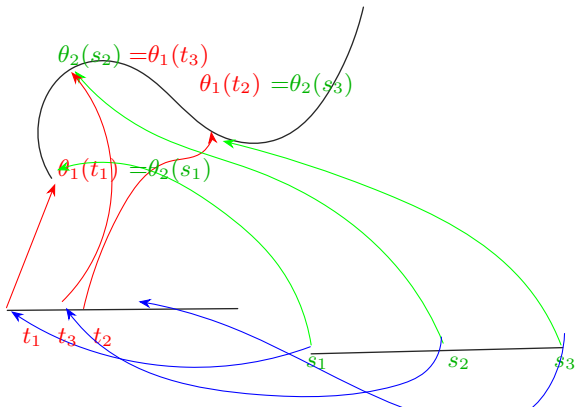
$$L(t_2) = \int_{t_1}^{t_2} \|\theta'(t)\| dt$$

So this is the formula for the arc length up to t_2

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

If we are given a non-unit speed parametrization θ

Consider a point p and $p = \theta(t_1) = \theta_2(t_2)$

t_2 is the arc length of the curve until $p = \theta(t_1)$

How would you define the arc length of the curve between the points $\theta(t_1)$ and $\theta(t_2)$

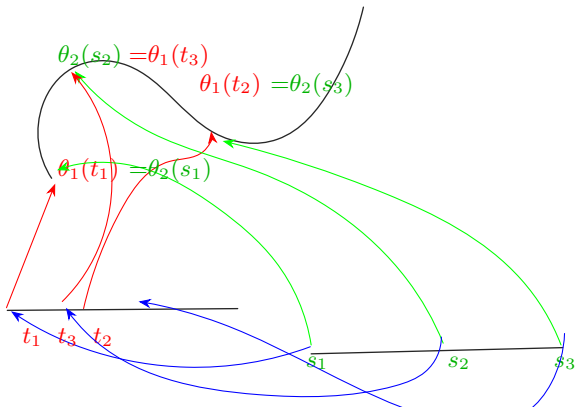
$$L(t_2) = \int_{t_1}^{t_2} \|\theta'(t)\| dt$$

Here we are assuming that the starting t_1 is fixed, and we are trying to figure out the distance upto a given t_2

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

If we are given a non-unit speed parametrization θ

Consider a point p and $p = \theta(t_1) = \theta_2(t_2)$

t_2 is the arc length of the curve until $p = \theta(t_1)$

How would you define the arc length of the curve between the points $\theta(t_1)$ and $\theta(t_2)$

$$L(t_2) = \int_{t_1}^{t_2} \|\theta'(t)\| dt$$

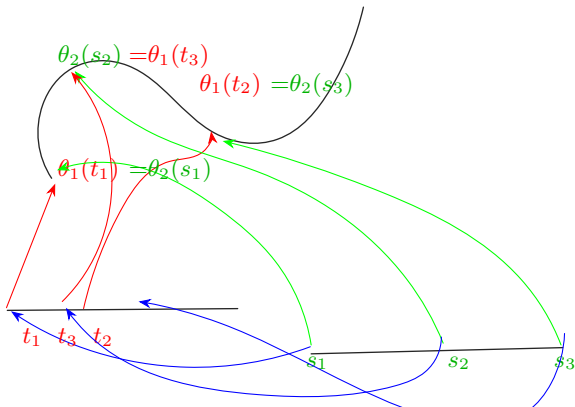
Exercise

Here is an exercise for you to think about

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



Think about why we can convert any parametrization into a unit speed one

The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

If we are given a non-unit speed parametrization θ

Consider a point p and $p = \theta(t_1) = \theta_2(t_2)$

t_2 is the arc length of the curve until $p = \theta(t_1)$

How would you define the arc length of the curve between the points $\theta(t_1)$ and $\theta(t_2)$

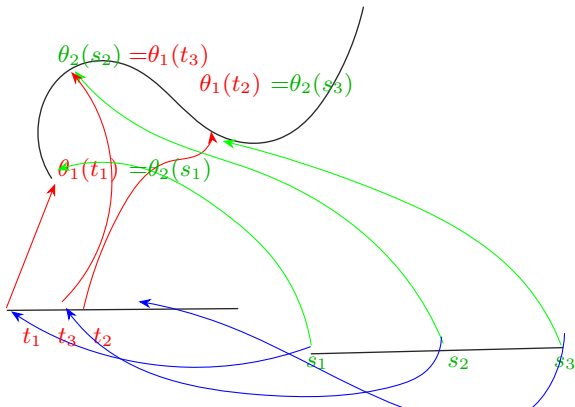
$$L(t_2) = \int_{t_1}^{t_2} \|\theta'(t)\| dt$$

Exercise Given a parametrization θ , show that $\theta(L^{-1}(t))$ is a unit speed parametrization

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



by first applying the *inverse* of the arc length formula on the parameter

The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

If we are given a non-unit speed parametrization θ

Consider a point p and $p = \theta(t_1) = \theta_2(t_2)$

t_2 is the arc length of the curve until $p = \theta(t_1)$

How would you define the arc length of the curve between the points $\theta(t_1)$ and $\theta(t_2)$

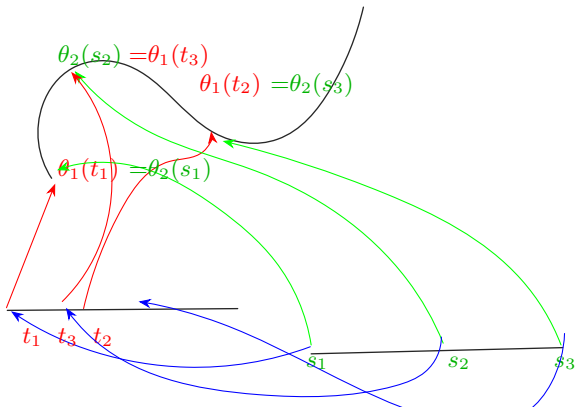
$$L(t_2) = \int_{t_1}^{t_2} \|\theta'(t)\| dt$$

Exercise Given a parametrization θ , show that $\theta(L^{-1}(t))$ is a unit speed parametrization

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



If you are familiar with the formula for centripetal acceleration in physics

The parameter in a unit speed parametrization coincides with the arc length

If θ_2 is a unit speed parametrization, from (a, b)

The parameter t matches with the arc length up to $\theta_2(t)$

If we are given a non-unit speed parametrization θ

Consider a point p and $p = \theta(t_1) = \theta_2(t_2)$

t_2 is the arc length of the curve until $p = \theta(t_1)$

How would you define the arc length of the curve between the points $\theta(t_1)$ and $\theta(t_2)$

$$L(t_2) = \int_{t_1}^{t_2} \|\theta'(t)\| dt$$

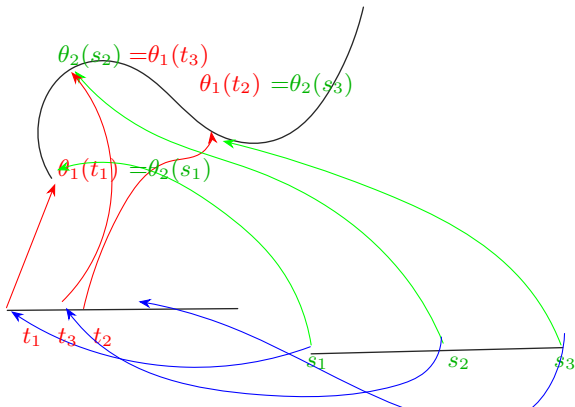
Exercise Given a parametrization θ , show that $\theta(L^{-1}(t))$ is a unit speed parametrization

Exercise Consider a circle of radius r

Can every parametrization be changed to a unit speed one?

$$\theta(t) = (\cos(e^t), t^3 + \sin(t^2 + e^{t^2}))$$

Goal: Find a definition of curvature that can use any parametrization



try to prove it using the techniques we have learned here!

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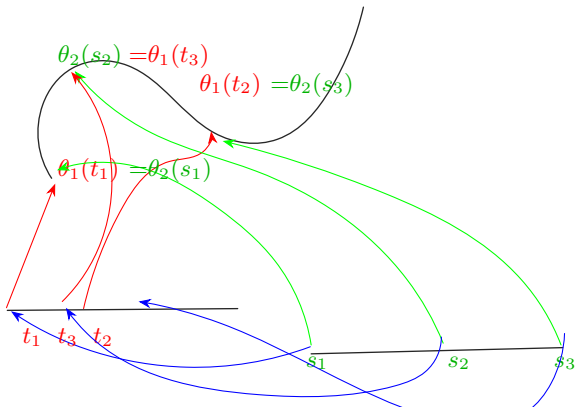
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If something is moving in a circle with velocity ν , you can choose a parametrization to model that

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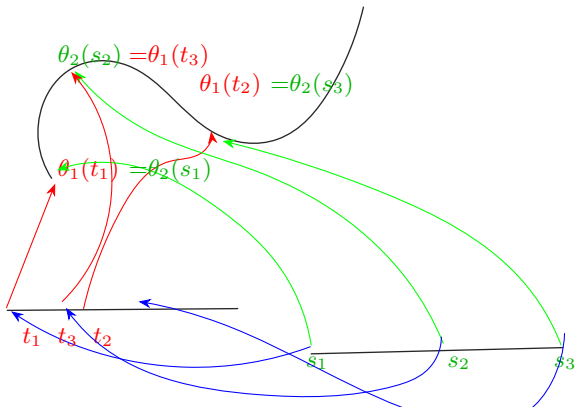
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Exercise Consider a circle of radius r . Find a parametrization for it which has speed ν .

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Then simply compute the acceleration

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